

Module 04 Lab - Exploratory Data Analysis **Objective:** To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model. **This lab is fully coded.** Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading ****What is Exploratory Data Analysis (EDA)?**** EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions. We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Seaborn is a library built on top of Matplotlib that makes creating beautiful plots easier.

# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

print("--- First 5 Rows ---")
print(df.head())
print("--- Basic Info ---")

# .info() is a great first command. It tells us the column names, how many non-null values are in each column, and the
```

```
--- First 5 Rows ---
  PassengerId  Survived  Pclass \
0             1         0       3
1             2         1       1
2             3         1       3
3             4         1       1
4             5         0       3

                                Name    Sex  Age  SibSp  \
0                Braund, Mr. Owen Harris  male  22.0     1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0     1
2      Heikkinen, Miss. Laina                female  26.0     0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0     1
4      Allen, Mr. William Henry                male  35.0     0

   Parch    Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.2500   NaN       S
1      0   PC 17599  71.2833   C85       C
2      0  STON/O2. 3101282  7.9250   NaN       S
3      0    113803  53.1000  C123       S
4      0   373450   8.0500   NaN       S

--- Basic Info ---
```

Part 2: Descriptive Statistics Let's start by getting a high-level numerical summary of the data.

The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columns
```

```
print("---- Descriptive Statistics ----")
print(df.describe())
print("---- Key Insights from Statistics ----")
print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")
print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
print(f"Fares ranged from ${df['Fare'].min()} to a whopping ${df['Fare'].max()}.")
```

```
---- Descriptive Statistics ----
      PassengerId  Survived  Pclass     Age  SibSp  \
count    891.000000    891.000000    891.000000  714.000000  891.000000
mean     446.000000     0.383838     2.308642   29.699118     0.523008
std      257.353842     0.486592     0.836071   14.526497     1.102743
min        1.000000     0.000000     1.000000     0.420000     0.000000
25%      223.500000     0.000000     2.000000   20.125000     0.000000
50%      446.000000     0.000000     3.000000   28.000000     0.000000
75%      668.500000     1.000000     3.000000   38.000000     1.000000
max      891.000000     1.000000     3.000000   80.000000     8.000000
```

```
      Parch  Fare
count    891.000000  891.000000
mean         0.381594  32.204208
std         0.806057  49.693429
min          0.000000   0.000000
25%          0.000000   7.910400
50%          0.000000  14.454200
75%          0.000000  31.000000
max          6.000000  512.329200
```

```
---- Key Insights from Statistics ----
The average age of a passenger was 29.7 years.
The overall survival rate was 38.4%.
Fares ranged from $0.0 to a whopping $512.3292.
```

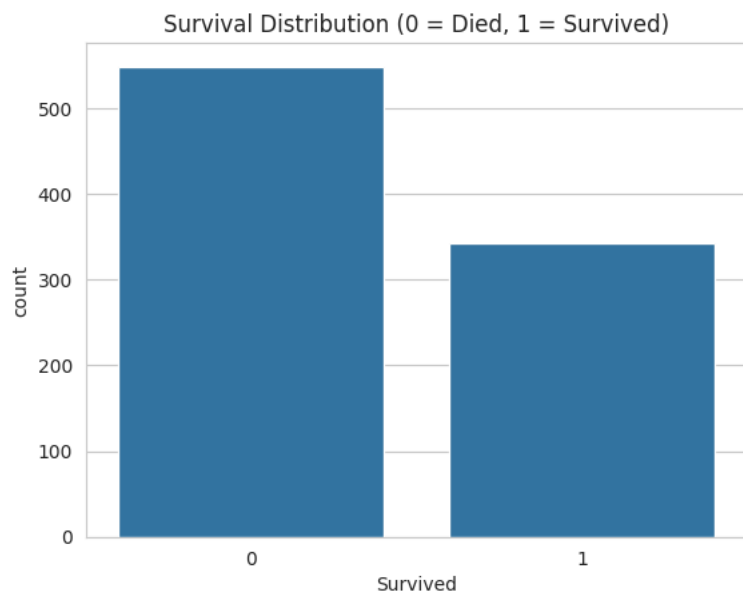
Part 3: Visual EDA - Telling Stories with Plots Numbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights. **A Note on**

- ✓ **Plotting Libraries:**
 - * **Matplotlib:** The foundational library, gives you full control over everything.
 - * **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

- ✓ Visualization 1: How many survived? A simple `countplot` is the best way to see the distribution of a categorical variable, like our target `Survived`.

```
sns.set_style('whitegrid')
# Sets a nice visual style for our plots plt.figure(figsize=(8, 6))

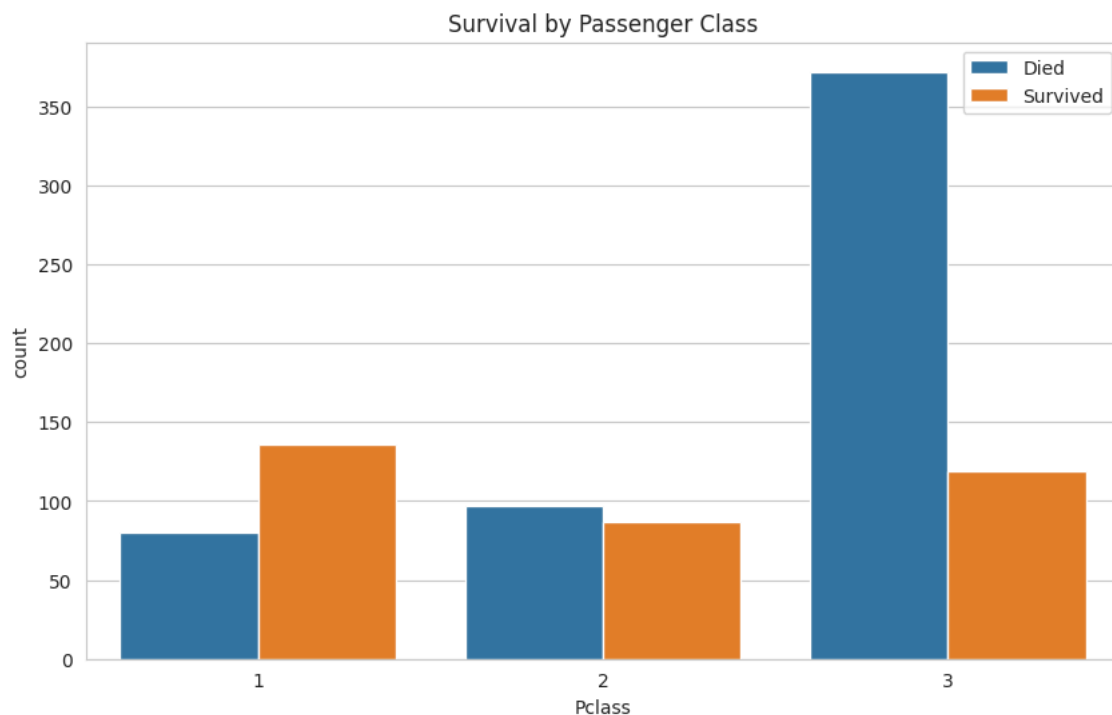
sns.countplot(x='Survived', data=df)
plt.title('Survival Distribution (0 = Died, 1 = Survived)')
plt.show()
print("Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes
```



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Visualization 2: Does passenger class matter for survival? Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to
```

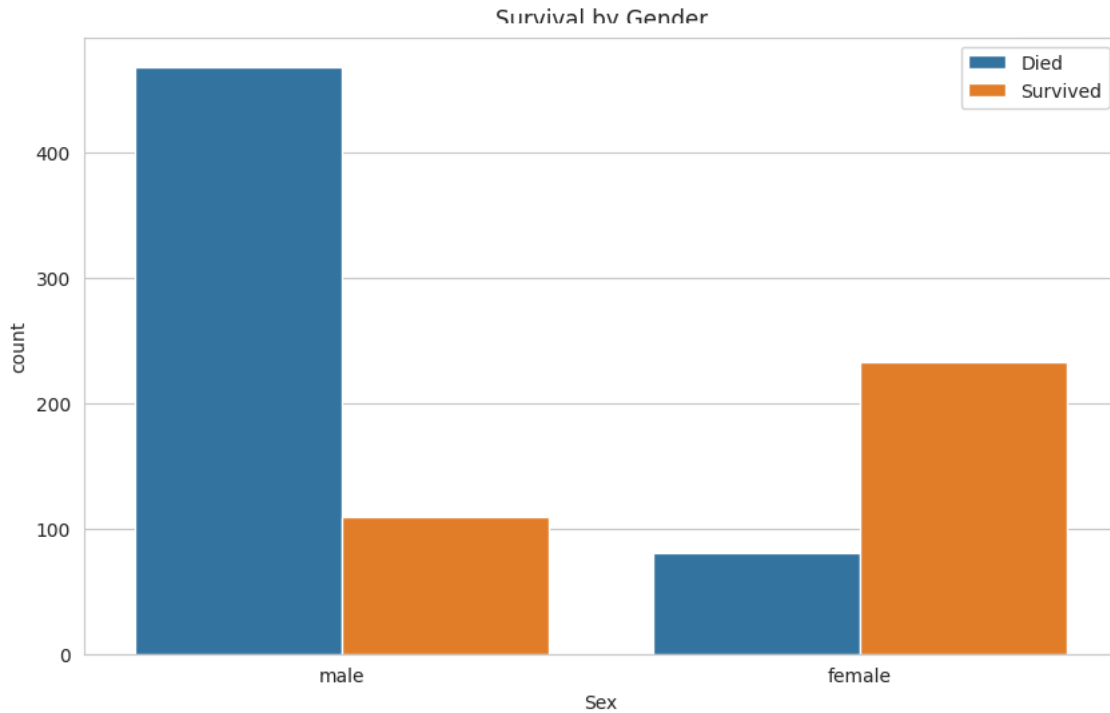


Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd cla

Visualization 3: What about gender? The 'women and children first' mantra is famous. Let's see if the data supports it.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Sex', hue='Survived', data=df)
```

```
plt.title('Survival by Gender')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is a
```



Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another v

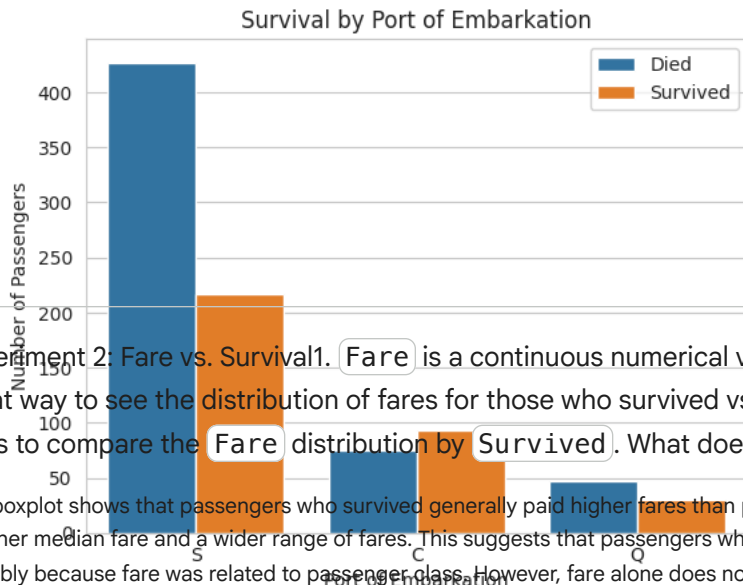
- Visualization 4: How does age play a role? For a continuous variable like `Age`, a histogram is a great way to see its distribution.

```
# A FacetGrid allows us to create multiple plots side-by-side to compare distributions. # Here, we create one histogr
```

- Part 4: Student Experimentation **Instructions:** Create your own visualizations to explore other relationships in the data.

- Experiment 1: Port of Embarkation1. The `Embarked` column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton).2. Create a `countplot` to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

```
# Experiment Port of Embarkation
sns.countplot(data=df, x='Embarked', hue='Survived')
plt.title('Survival by Port of Embarkation')
plt.xlabel('Port of Embarkation')
plt.ylabel('Number of Passengers')
plt.legend(['Died', 'Survived'])
plt.show()
```

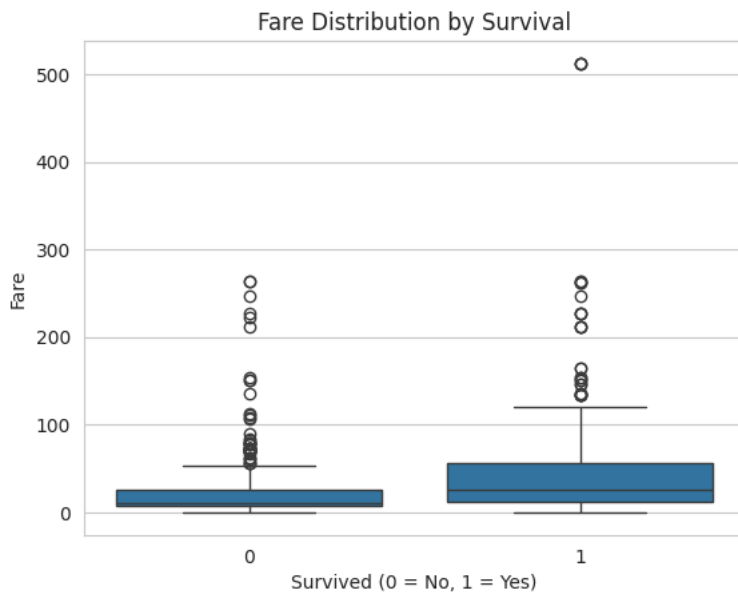


Experiment 2: Fare vs. Survival. 1. Fare is a continuous numerical variable. 2. A boxplot or violinplot is a great way to see the distribution of fares for those who survived vs. those who didn't. 3. Create one of these plots to compare the Fare distribution by Survived. What does it tell you?

The boxplot shows that passengers who survived generally paid higher fares than passengers who did not survive. The survivor group has a higher median fare and a wider range of fares. This suggests that passengers who paid higher fares had a greater chance of survival, possibly because fare was related to passenger class. However, fare alone does not determine whether someone survived.

```
# Experiment 2: Fare vs. Survival
```

```
sns.boxplot(data=df, x='Survived', y='Fare')
plt.title('Fare Distribution by Survival')
plt.xlabel('Survived (0 = No, 1 = Yes)')
plt.ylabel('Fare')
plt.show()
```



Knowledge Check Instructions: Answer the following questions in this markdown cell. 1.

****What is the primary goal of Exploratory Data Analysis (EDA)?**

The main goal of EDA is to understand the data before building a model. It helps us find patterns, relationships, missing data, and anything unusual in the dataset.

****2. Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic?** (Describe them in terms of class, gender, and age). Based on the plots, a female passenger in first class who was younger had a better chance of surviving.

3. Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't? Visualizing the data makes it easier to see what is really going on. A graph can show patterns, differences,